CONSCIENTIOUSNESS AND LABOUR MARKET OUTCOMES: A NIDS ANALYSIS

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Abstract

Abstract to be written here. The abstract should not be too long and should provide the reader with a good understanding what you are writing about. Academic papers are not like novels where you keep the reader in suspense. To be effective in getting others to read your paper, be as open and concise about your findings here as possible. Ideally, upon reading your abstract, the reader should feel he / she must read your paper in entirety.

1. Introduction

2. Preliminary OLS

Conscientiousness plays a multitude of roles in the formation of various outcomes in an individual's life. Some of these outcomes include educational choice, behavioural decisions, wages, and others. To address issues of reverse causality, education and conscientiousness were modelled separately to determine if a causal relationship between the two variables was present.

Endogeneity issues arise through concerns of reverse causality between education and conscientiousness, as well as measurement error of the conscientiousness variable. As the table below demonstrates, conscientiousness has a significant positive role in the attainment of education. After the inclusion of control variables, this relationship remains significant. Appendix B lays out the result from the regression of education on conscientiousness, showing that the former is also positively correlated with the latter. This result strengthens the need for a method which accounts for this endogeneity introduced in the data generating process for wages. Allowing for this relationship to be accounted for through a simultaneous equation structural modelling approach, endogeneity issues can be reduced. Measurement error occurs as a result of the questions making up the variable of conscientiousness being imperfect proxies for the latter. By simulating factor scores of this measure through structural modelling simulation, more accurate scores are obtainable.

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Table 1: Linear model results of conscientiousness on education

	Dependent variable: Education	
Variables	(1)	(2)
Constant	12.062***	11.943***
	(0.052)	(0.094)
Conscientiousness	0.303***	0.367***
	(0.049)	(0.060)
Female		0.695***
		(0.126)
Coloured		-0.755***
		(0.177)
Asian/Indian		0.933
		(0.544)
White		1.126*
		(0.498)
Wealth		0.452***
		(0.042)
N	5062	3516
R-squared	0.007	0.066
Adj. R-squared	0.007	0.065

Note: *** p<0.001, ** p<0.01, * p<0.05. Standard errors in parentheses.

To investigate the presence of a linear relationship between wages and conscientiousness, and employment status and conscientiousness a linear model was constructed. Results thereof are presented below.

Table 2: Linear model results of conscientiousness on employment status and wages

	Dependent variable: Employment status = 1					Dependent variable: *Log*(Wages)				
Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Constant	0.474***	-2.300***	-2.338***	-1.910***	-1.910***	7.422***	1.956*	2.162*	3.330**	3.330**
	(0.008)	(0.339)	(0.331)	(0.382)	(0.382)	(0.019)	(0.895)	(0.886)	(1.025)	(1.025)
Conscientiousness	0.045***	0.035***	-0.001	0.004	0.004	0.156***	0.120***	0.017	0.022	0.022
	(0.007)	(0.007)	(0.007)	(0.008)	(0.008)	(0.018)	(0.017)	(0.018)	(0.021)	(0.021)
Education		0.027*	0.043***	0.048***	0.048***		0.102***	0.123***	0.077*	0.077*
		(0.011)	(0.011)	(0.013)	(0.013)		(0.026)	(0.027)	(0.033)	(0.033)
$Education^2$		-0.000	-0.001*	-0.001*	-0.001*		-0.001	-0.001	0.000	0.000
		(0.000)	(0.000)	(0.000)	(0.000)		(0.001)	(0.001)	(0.001)	(0.001)
Age		0.240***	0.238***	0.189***	0.189***		0.409***	0.390***	0.302**	0.302**
		(0.037)	(0.036)	(0.042)	(0.042)		(0.096)	(0.095)	(0.110)	(0.110)
Age^2		-0.005***	-0.005***	-0.004***	-0.004***		-0.009***	-0.009***	-0.006*	-0.006*
		(0.001)	(0.001)	(0.001)	(0.001)		(0.003)	(0.003)	(0.003)	(0.003)
Female			-0.210***	-0.200***	-0.200***			-0.578***	-0.573***	-0.573***
			(0.015)	(0.017)	(0.017)			(0.039)	(0.045)	(0.045)
Coloured			0.192***	0.208***	0.208***			0.295***	0.255***	0.255***
			(0.022)	(0.024)	(0.024)			(0.054)	(0.059)	(0.059)
Asian/Indian			0.009	0.121	0.121			0.595**	0.719**	0.719**
			(0.072)	(0.079)	(0.079)			(0.211)	(0.227)	(0.227)
White			0.339***	0.348***	0.348***			1.064***	0.697***	0.697***
			(0.056)	(0.071)	(0.071)			(0.170)	(0.199)	(0.199)
Wealth				0.002	0.002				0.087***	0.087***
				(0.006)	(0.006)				(0.016)	(0.016)
N	4269	4260	4260	3238	3238	3284	3276	3099	2308	2308
R-squared	0.009	0.086	0.142	0.135	0.135	0.023	0.136	0.219	0.235	0.235
Adj. R-squared	0.009	0.084	0.140	0.133	0.133	0.022	0.134	0.217	0.231	0.231

Note: *** p<0.001, ** p<0.01, * p<0.05. Standard errors in parentheses (robust standard errors reported in columns (1)-(5)).

References

Appendix

Appendix A: Conscientiousness PCA loadings

Variables	PC1
Exercise	-0.682
Emotional hope	-0.630
Planning tertiary education	-0.122
Smoked	0.351

Note: Obtained from PCA results

Appendix B: Regressing education on conscientiousness

Table 3: Linear model results of education on conscientiousness

	Dependent variable: Conscientiousness	
Variables	(1)	(2)
Constant	-0.299***	-0.074
	(0.050)	(0.062)
Education	0.024***	0.029***
	(0.004)	(0.005)
Female		-0.610***
		(0.034)
Coloured		0.258***
		(0.049)
Asian/Indian		0.155
		(0.152)
White		0.625***
		(0.139)
Wealth		0.026*
		(0.012)
N	5062	3516
R-squared	0.007	0.109
Adj. R-squared	0.007	0.107

Note: *** $p<0.001$, ** $p<0.01$, * $p<0.05$. Standard errors in parentheses.